

Annotation of Face Effects as a Tool for Valence Analysis

Valence Asymmetries in Expressive Language

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VALENCE
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I want you to be happy (1)



I want you to be happy (2)



What is happening here?

- The utterance *I want you to be happy* has a **different valence** in (1) and (2).
- **Valence**: the hedonic quality of an affective experience (i.e. whether the experience is pleasant or unpleasant, judged as good or bad, etc.) (Ortony et al. 1988, Barrett 1996, Barrett 2006)

In (1) it has a **negative valence** - as experienced by Yaz

In (2) it has a **positive valence** - ”

We can intuit this from Yaz’s reaction to the utterance, and the context in general.

Puzzle: disambiguating the valence of utterances

- Given a dialogue, we have intuitions about what expressions have a negative or a positive valence within their context. We want to **capture these intuitions**.
- **Analyzing interactions in terms of face** is useful in deriving how we interpret an utterance, including **disambiguating its valence**.
- Goal: develop an annotation system of the perlocutionary effect on face for application on interactional data

to **establish a link between face effects and the valence of an utterance**.

Annotation: what is it good for?

- Annotation is a link between the theory and its application.
 - Can only be developed if the underlying **theory** is properly worked out.
 - Defines what questions can be asked through its **application**.
- With it, we can study **the role of face effects in interaction** (relatively unexplored so far) via annotation of data and/or recreating behaviours.
- Can yield insights about related phenomena: e.g. valence, politeness, confrontation, social norms, etc.

Structure of the talk

1. Introduction
2. Theoretical background
3. Proposed revisions and contributions
4. Annotation methodology
5. Assessing of the annotation system
6. Challenges and future work

Theoretical Background

The role of affective expression in interaction

- Affect (feelings, moods, dispositions, etc.) is expressed through all levels of language: verbal, prosodic, kinesic. (Ochs & Schieffelin 1989, Peräkylä 2012)
- Affective expression shapes and is shaped by interaction (Sorjonen & Peräkylä 2012, Peräkylä 2012)
- Affective expression can serve social functions, e.g. building the social identity (Ochs 1996)

Face: what and where

- **Face** is a notion that aims to represent some aspect of this social identity:

“The public self-image that every member of a society wants to claim for themselves.” (Brown & Levinson 1987)

- Components of speakers' internal state:

- Beliefs
- Commitments
- etc.
- **Face**

Face in interaction [Brown & Levinson 1987]

- B&L assume that interactants:
 - 1) have face (and are mutually aware of face sensitivities)
 - 2) reason about ways to achieve ends
- Therefore, face concerns constrain actions
- **Face is constantly being managed in interaction**
 - Politeness strategies are applied to actions so that face concerns are attended to (i.e. face threat is minimized)

Face: two aspects [Brown & Levinson 1987]

Negative face Freedom of action and freedom from imposition.



The want that one's actions be unimpeded by others.

Positive face Positive self-image, being appreciated and approved of by others.



The want that one's wants be desirable to others.

Classification of Face Threatening Acts [Brown & Levinson 1987]

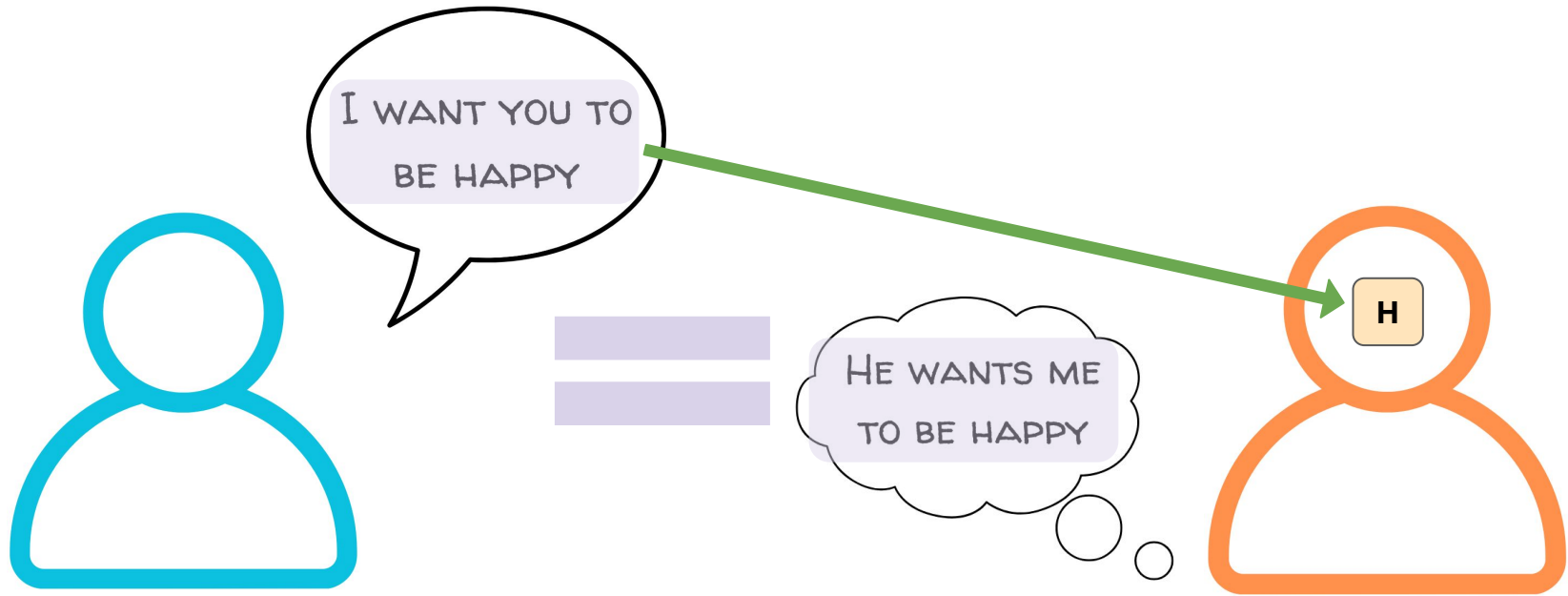
- Some acts are *intrinsically* face threatening
 - **Because B&L assume that interlocutors correctly recognize others' intentions.**

Meaning of an utterance in interaction

(Austin 1962)

- Locution - what is said
- Illocution - what is intended
- Perlocution - what happens as a result

Inconsistencies in Brown & Levinson's approach



- Therefore intention (illocutionary force) determines whether face is threatened

Proposed Revisions and Contributions

1. Focus on perlocution

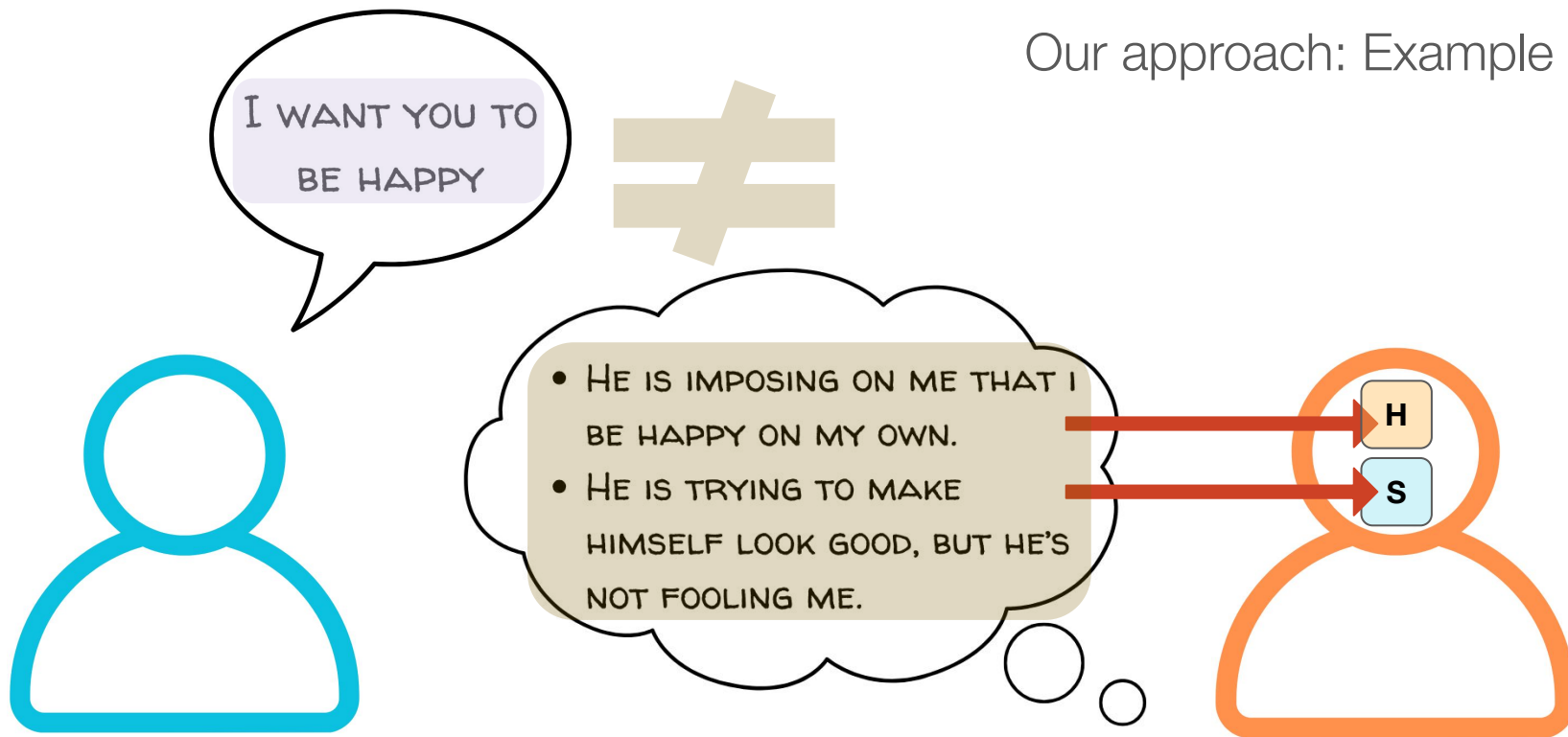
- No act/utterance can be *intrinsically* face threatening. The hearer must interpret it as such.
- Face threat is not determined by the speaker's intention. S's intention may be the same in (1) and (2).
- But S's intention is not always decoded the same way by H (i.e. **intention does not necessarily match the effect**).



VS.

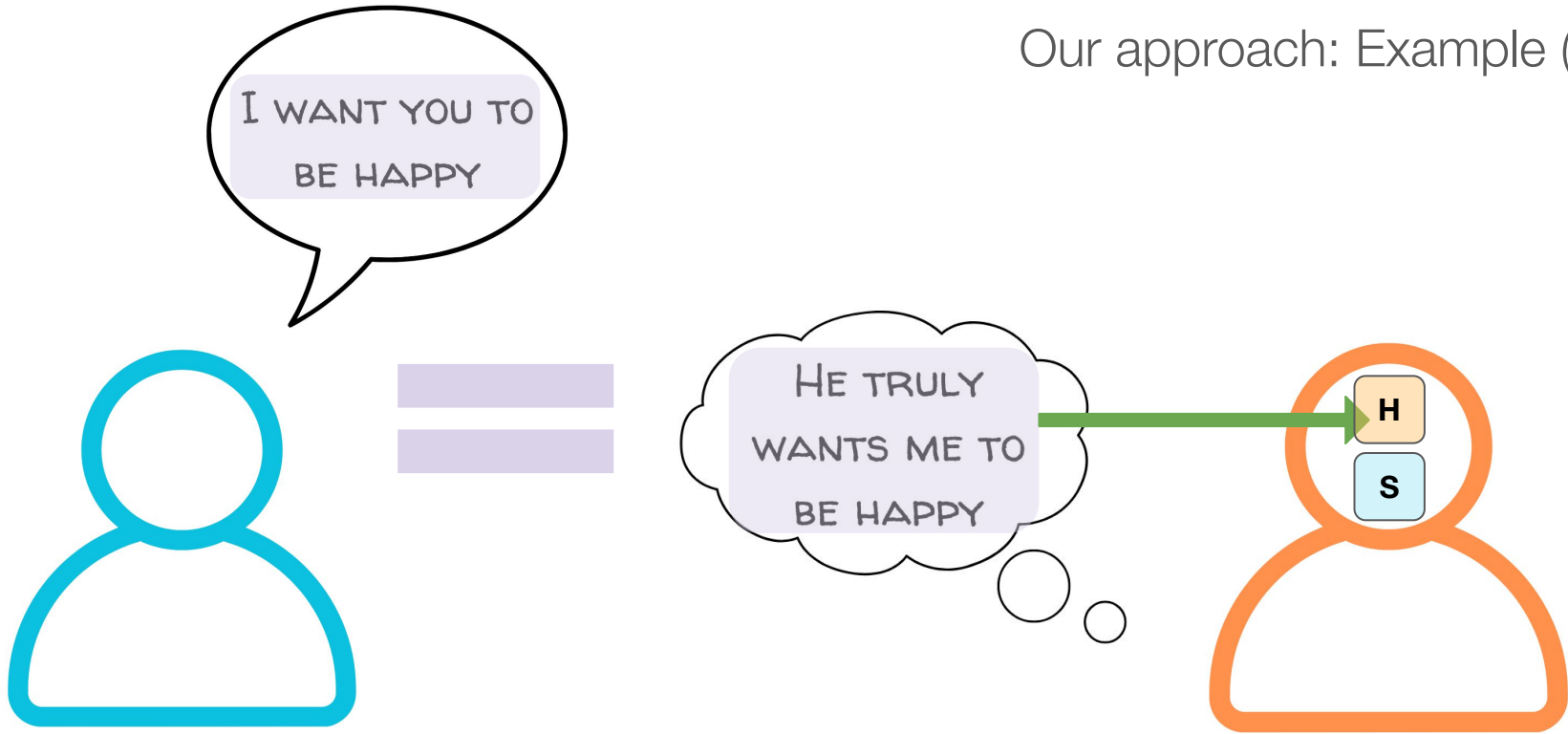


Our approach: Example (1)



- Face effects happen after interpretation

Our approach: Example (2)



2. Face raising

- If it is possible to mitigate face threat (with politeness strategies), then there is such a thing as a symmetrically opposing effect: **face raising**.



Positive face: the want that one's wants be desirable to others.

3. One act, multiple face effects

- One act can have multiple effects.

E.g.: *Shut up!*

- **Threatens (H's) negative face**: H's freedom of action is imposed on
- **Threatens (H's) positive face**: H's opinion is not valued

Summary of our contributions



- Focus on **perlocution** rather than illocution: face effects of an act are anchored to the reaction (which includes an in/correct reconstruction of S's intentions)
- We consider not only face threatening but also **face raising** effects
- We acknowledge that one act may have **multiple effects** on face

I want you to be happy 👍 / 👎

Back to the puzzle:

How do we disambiguate the valence of an expression in different contexts?

- Now that the theory has been revised
- we can develop an applicable annotation system as a tool to disambiguate valence

Annotation Methodology

Desired properties

We hope the methodology to be some balance of the following:

- **Simple** - Possible to train annotators with no prior theoretical background.
- **Consistent** - Generally good agreement (Cohen's Kappa > 0.4 , ideally > 0.6)
- **Informative** - The extracted information allows researchers to study and answer a broad set questions regarding language use.

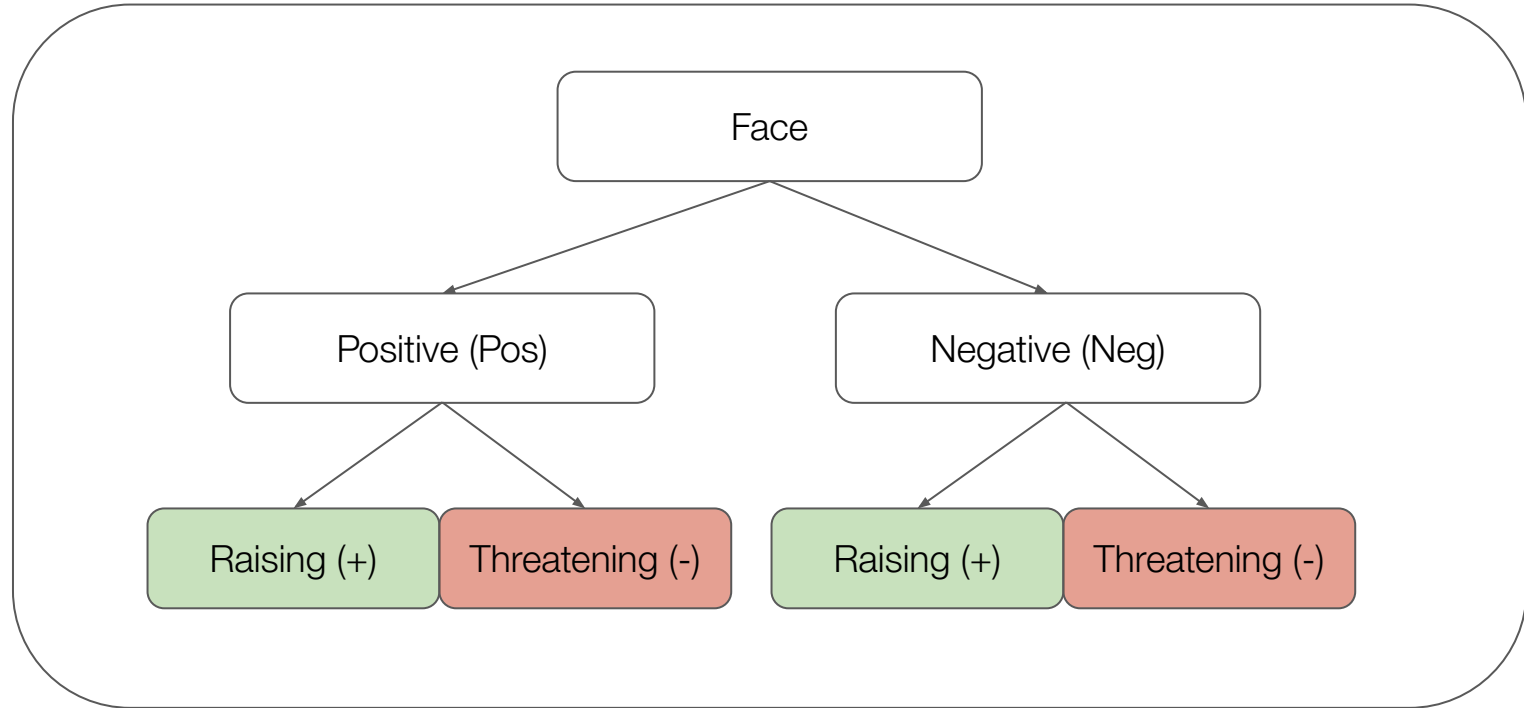
Components of an annotation

In our schema a *face effect* contains the following information:

- (1) Speaker - The (specific) person whose utterance caused the face effect.
- (2) Hearer - The (specific) person who is being affected by the face effect.
- (3) Source Utterance - Utterance which initiated this face effect.

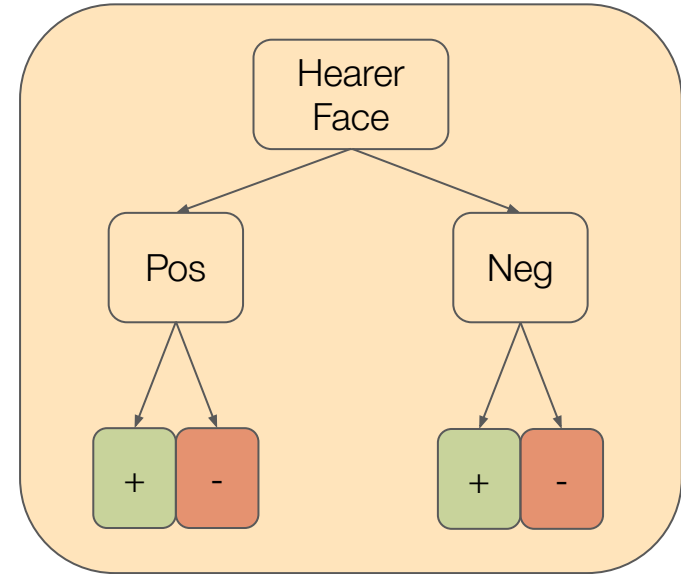
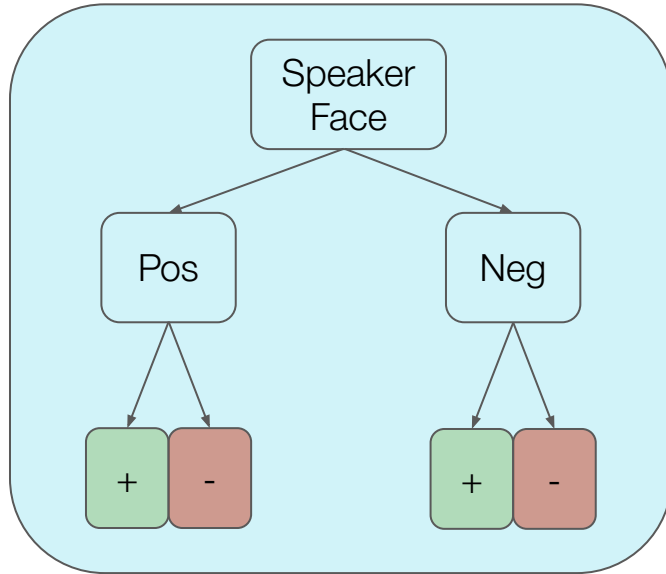
- | | | |
|--|---|--------------------|
| <ul style="list-style-type: none">(4) Speaker / Hearer Orientation (S / H)(5) Positive / Negative Face Orientation (Pos / Neg)(6) Raising / Threatening Orientation (+ / -) | } | 8 Possible Effects |
|--|---|--------------------|

Representation of face

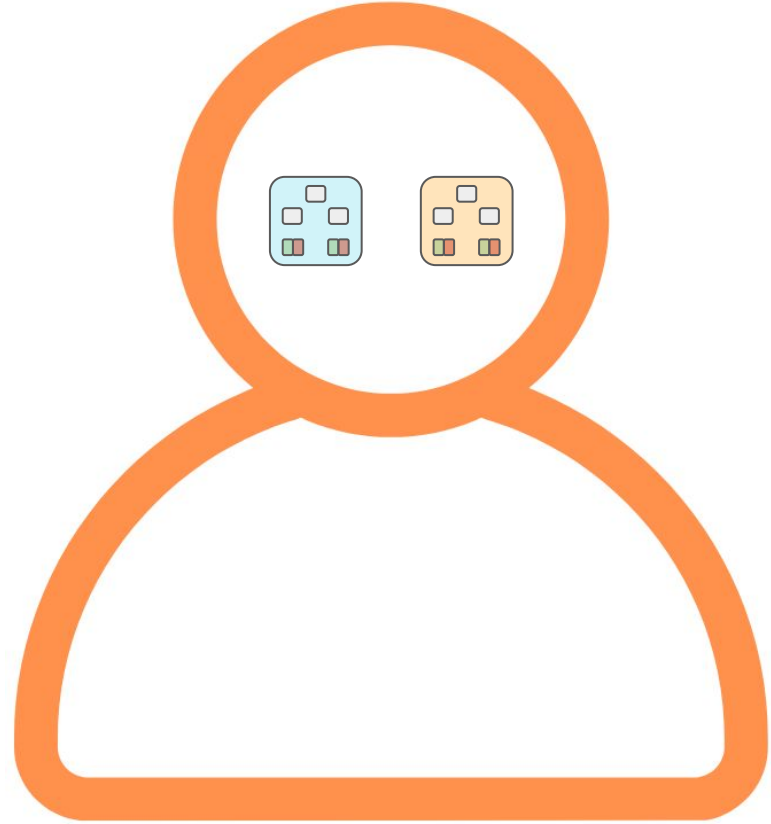
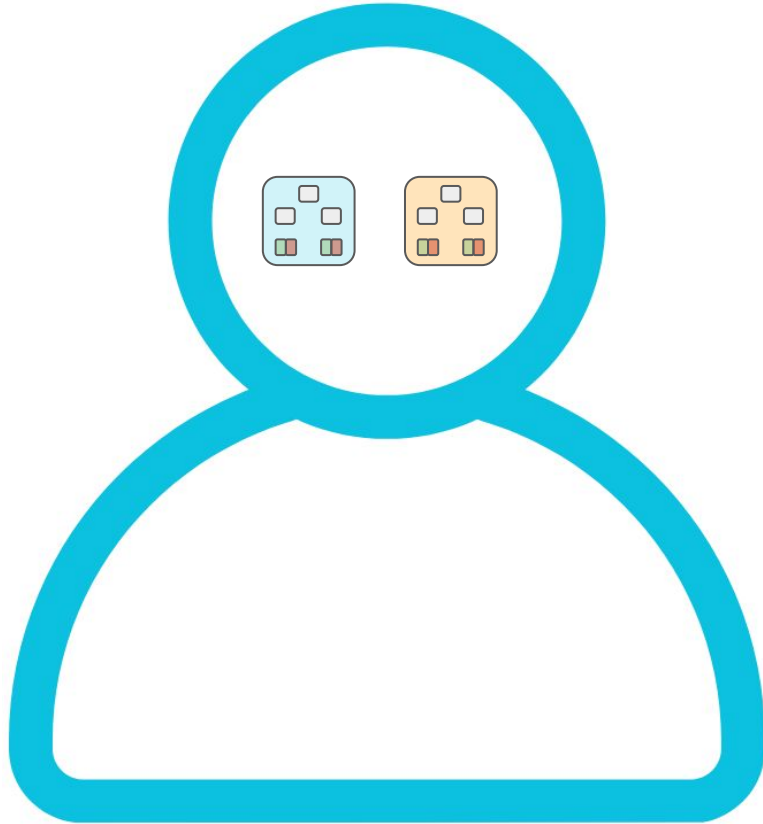


Representation of face

(one interlocutor's internal state in a dyadic dialogue)



Repres. of face (both interlocutors in a dyadic dialogue)



Annotation workflow

- (1) Read the source utterance and reacting turn
- (2) For each of the 8 possible face effects, identify presence individually
 - Refer to the guidelines in the Annotation Manual for what constitutes presence
 - The definition is less specific but most important
 - When in doubt, abide by the criteria
- (3) Revise if context reveals the effect differs from what was first annotated

Annotation Manual entry example

[HPos+] Hearer Positive Face Raising

Definition

A discourse action which causes H to believe that another (possibly S) thinks of H similarly to how H would likely wish.

Criteria

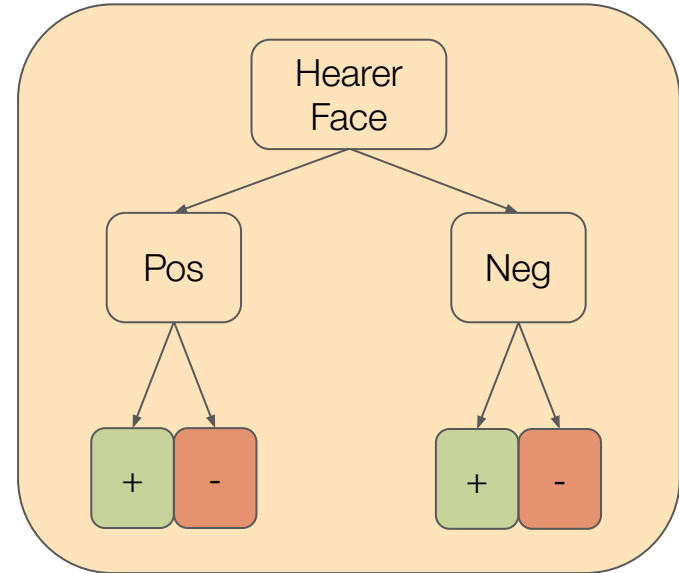
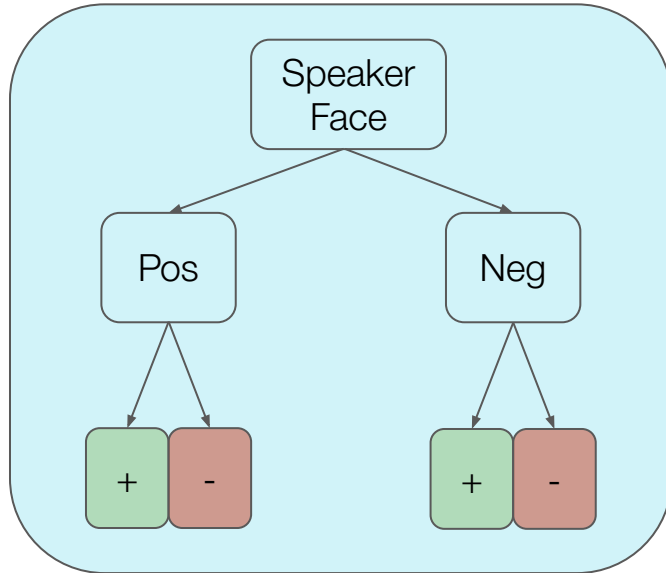
1. Those acts that show that S has a positive evaluation of some aspect of H's positive face:
 - (a) expressions of approval, compliments, admiration (S indicates that they find desirable one or more of H's wants, acts, qualities, possessions, beliefs or values)
 - (b) ...
2. Those that show that S cares about (or is aware of) H's positive face:
 - (a) ...

Annotation example

	A	B	C	D	E	F	G	H	I	J
1	Clip 1 [link]	SPos-	SNeg-	HPos-	HNeg-	SPos+	SNeg+	HPos+	HNeg+	Comment
2										
3	LCK: James "Lockie"									
4	YAZ: Yaz									
5										
6	LCK: the way I gone about things I said to you befo:re (0.4)									
7	LCK: was wrong. ((direct eye contact with Yaz))	☐	☐	☐	☒	☒	☐	☒	☐	
8	YAZ: [°(kay)°]	☐	☐	☐	☐	☐	☐	☒	☒	
9	LCK: [bu-] (0.4) we've been un unhappy for a long time	☐	☐	☒	☐	☐	☐	☒	☐	light HPos+ and
10	YAZ: °mm hm°=	☐	☐	☐	☐	☐	☐	☐	☒	
11	LCK: =you know, so .hhh	☐	☐	☐	☐	☐	☐	☐	☐	
12	LCK: I'm no- try- justify things?	☐	☐	☐	☐	☒	☐	☒	☐	
13	YAZ: ((raised eyebrows)) I just think the way that you (0.4)									
14	dealt with things (0.5)	☐	☐	☒	☐	☒	☐	☐	☐	
15	YAZ: I would never deal with it [in that way] cuz	☐	☐	☒	☐	☒	☐	☐	☐	
16	LCK: [I agree.]	☐	☐	☐	☒	☒	☒	☒	☐	
17	YAZ: the way I am with my morals is [I'm] very loyal girl?	☐	☐	☒	☐	☒	☐	☐	☐	
18	LCK: [mm]	☐	☐	☐	☐	☐	☐	☐	☒	
19	YAZ: [.hhh and] I would never've disrespected you:	☐	☐	☒	☐	☒	☐	☐	☐	
20	LCK: [°okay°]	☐	☐	☐	☐	☐	☐	☒	☒	
21	YAZ: la- ho:w do you think that's made me feel as a pe:rson [la-	☐	☐	☒	☒	☒	☐	☐	☐	
22	LCK: [yeah?]	☐	☐	☐	☐	☐	☒	☐	☐	
23	YAZ: ho:w d'you think that's [made my confidence] feel?	☐	☐	☒	☒	☒	☒	☐	☐	Is this SPos+? D
24	LCK: [no but see this]	☐	☐	☐	☐	☐	☒	☐	☐	Adil had HPos- :
25	LCK: okay	☐	☐	☐	☐	☐	☐	☒	☐	
26	YAZ: like [no]	☐	☐	☒	☒	☒	☒	☐	☐	this is the ultimate

Face effects annotation: *I want you to be happy*

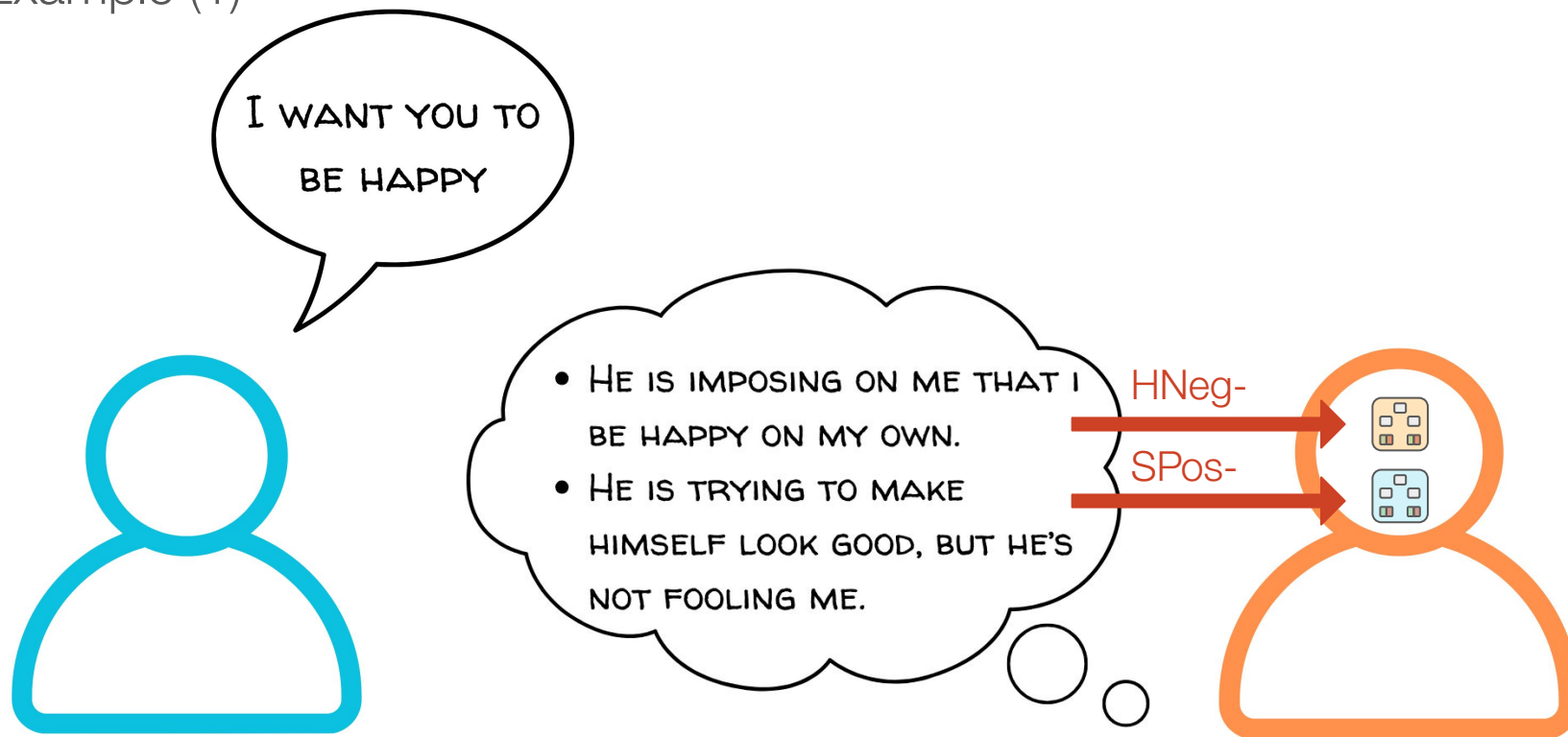
[H's internal state]



Hearer - The (specific) person who is being affected by the face effect.

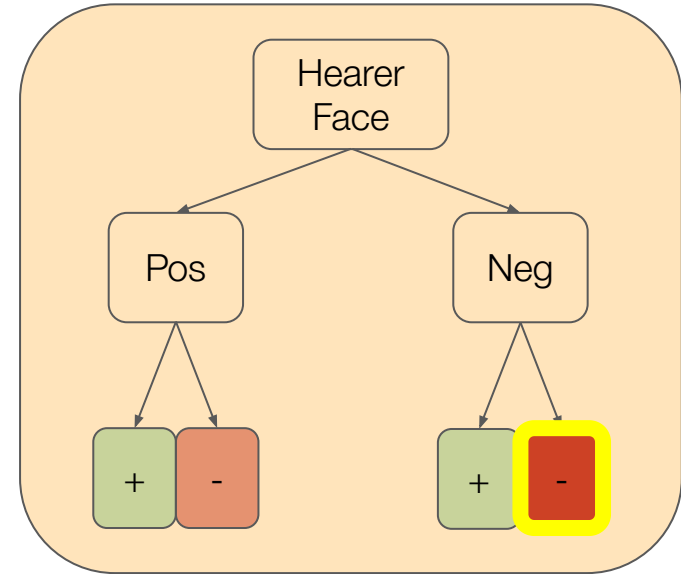
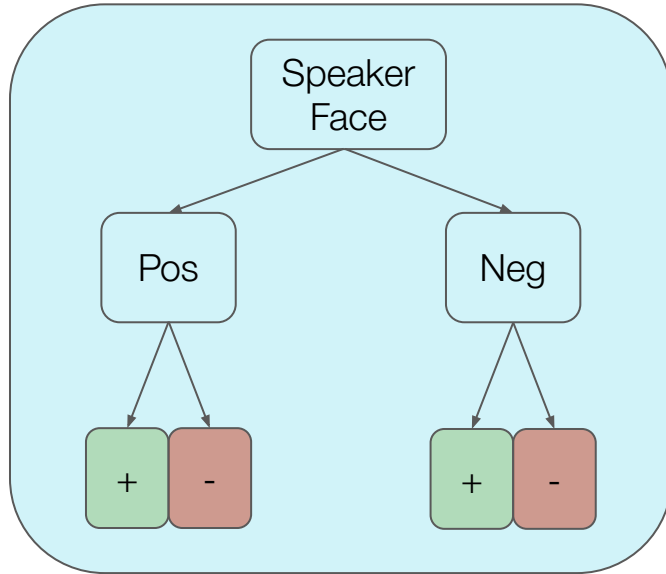
8 Possible Effects

Example (1)



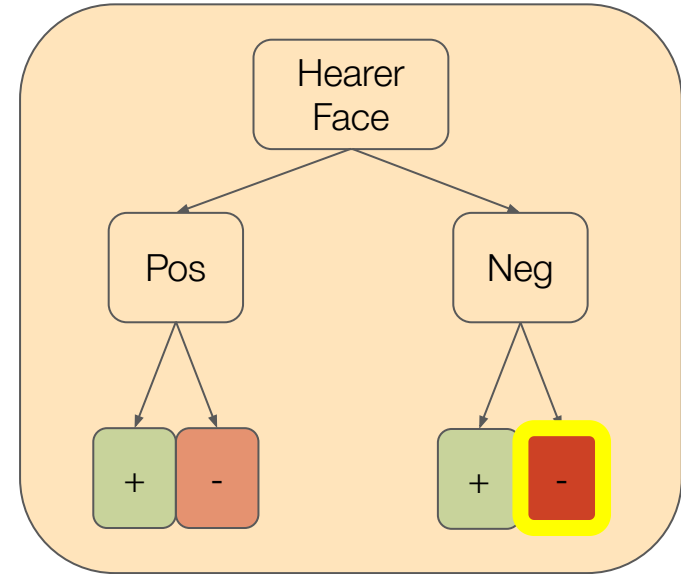
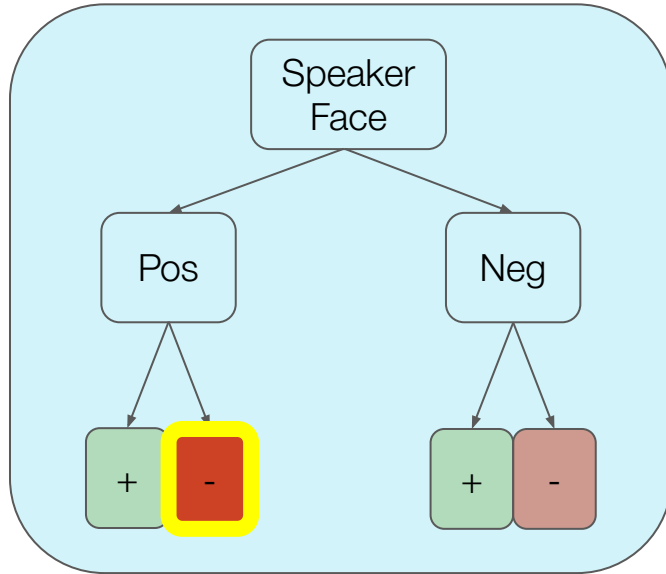
Example (1)

- HNeg- | Hearer Negative Face **Threatening**
 - A discourse action which causes H to believe that H is **more obligated** to act in a particular way.

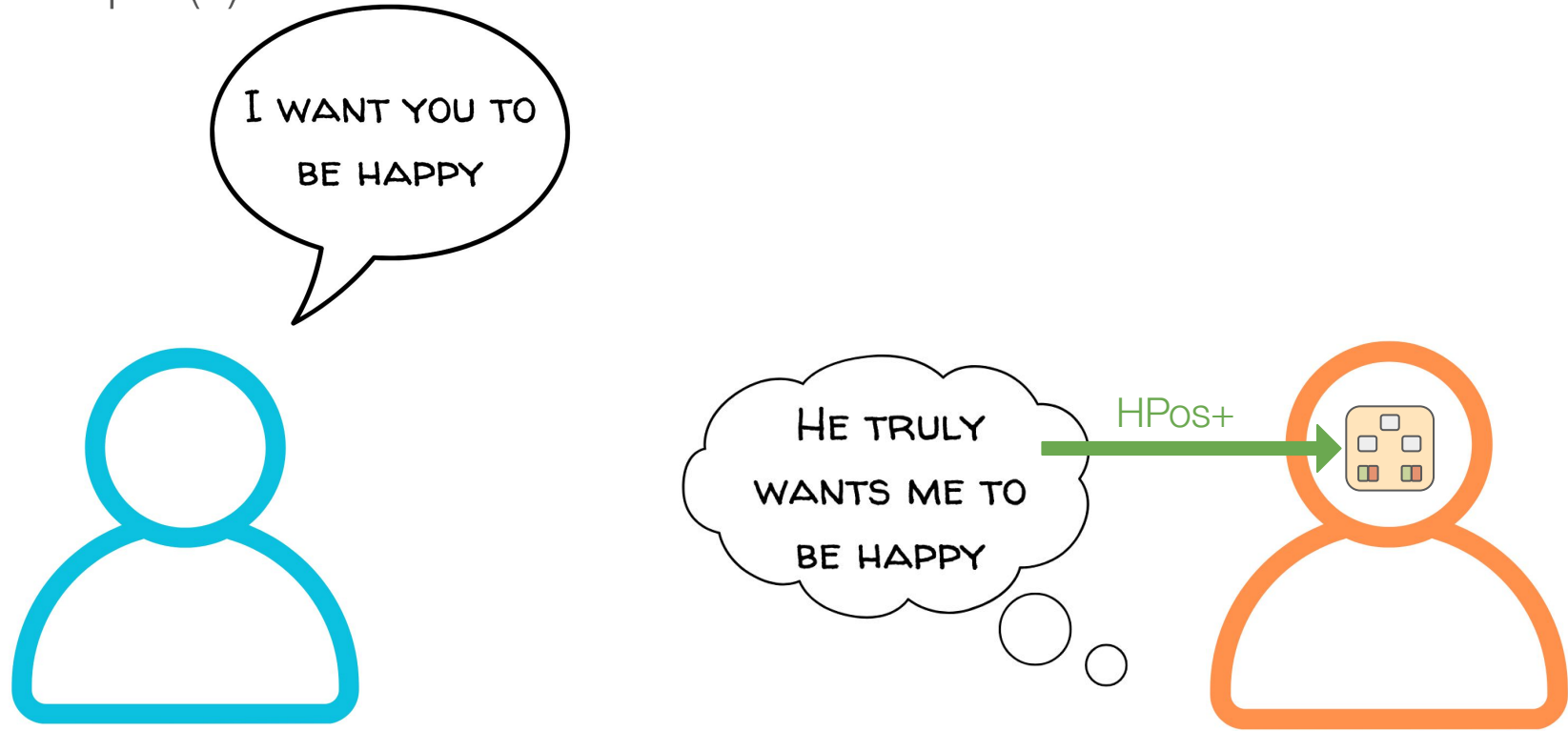


Example (1)

- SPos- | Speaker Positive Face **Threatening**
 - A discourse action which causes H (not necessarily the addressee) to think of S **differently** than S would likely wish.

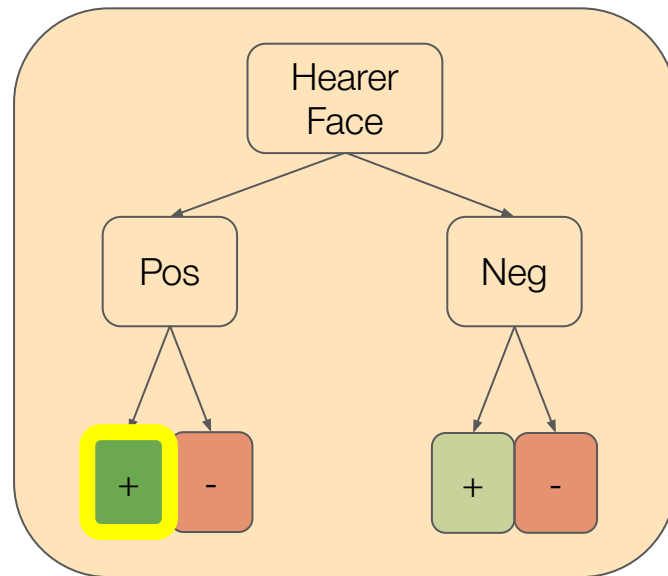
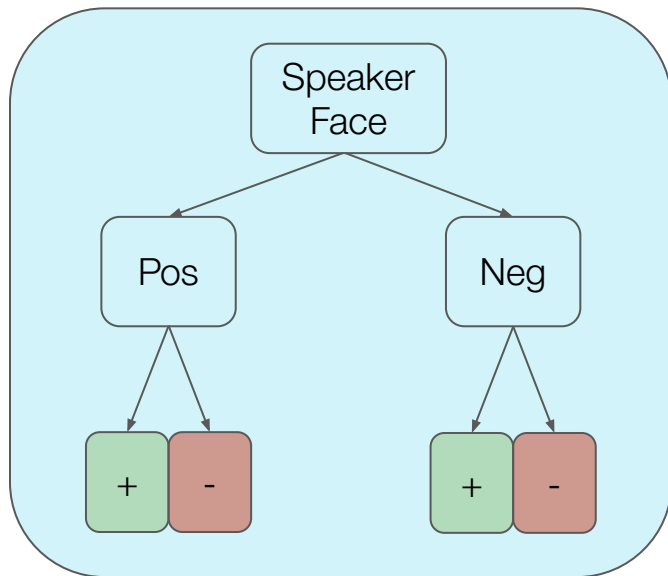


Example (2)



Example (2)

- HPos+ | Hearer Positive Face **Raising**
 - A discourse action which causes H to believe that another (possibly S) thinks of H **similarly** to how H would likely wish.



Annotation of (1) and (2)

(1)

Speaker	Utterance	Hearer	Face Effect(s)
LCK	<i>And I want you to be happy, I mean that from (the) [bottom of my heart], I want you to be happy.</i>	YAZ	HNeg-, SPos-
YAZ	<i>[((looks to the side)) Yeah... ((purses lips))</i>	LCK	

(2)

Speaker	Utterance	Hearer	Face Effect(s)
LCK	<i>I'm glad it's good, and I wish, I wish you all the best. I want you to be happy as well, you know that don't you!</i>	YAZ	HPos+
YAZ	<i>((smiling softly)) Yeah.</i>	LCK	

Annotation of (1) and (2)

(1)

Speaker	Utterance	Hearer	Face Effect(s)
LCK	<i>And I want you to be happy, I mean that from (the) [bottom of my heart], I want you to be happy.</i>	YAZ	HNeg-, SPos- Negative valence
YAZ	<i>(((looks to the side)) Yeah... ((purses lips))</i>	LCK	

(2)

Speaker	Utterance	Hearer	Face Effect(s)
LCK	<i>I'm glad it's good, and I wish, I wish you all the best. I want you to be happy as well, you know that don't you!</i>	YAZ	HPos+ Positive valence
YAZ	<i>((smiling softly)) Yeah.</i>	LCK	

Face effects and valence

- Preliminary claim:
Face effects are correlated with the valence of an expression.

Overall **face threat** → **negative valence**

Overall **face raising** → **positive valence**

- Caveat: weightiness is crucial in order to calculate total face effects but...



Further testing

- We can use annotation system of face effects to determine the valence of *some* utterances in their context.
- But how robust is this tool?

Assessing the Annotation System

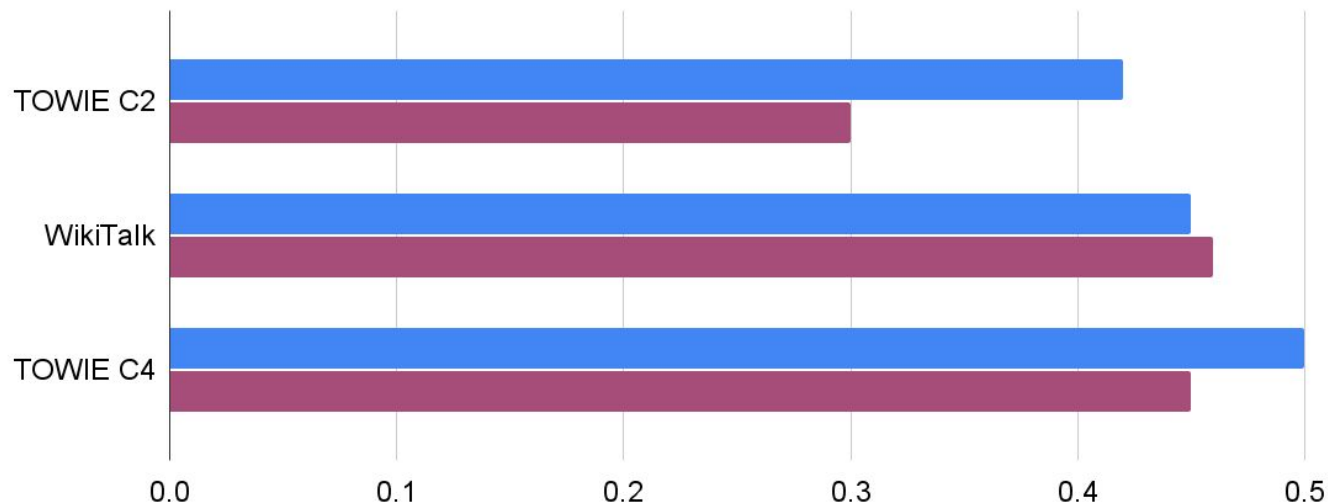
Pilot annotations to check inter-annotator agreement

Trial	Dialogue	Modality (Medium)	Speakers	Turns	Tone
1	TOWIE C2	Multimodal (Reality TV)	2	58	Confrontational
2	Wikipedia Talk Pages	Unimodal (Webforum thread)	5	48	Confrontational
3	TOWIE C4	Multimodal (Reality TV)	2	50	Variable

- Annotated separately (Adil and Sara)
- With discussion and revision of the Annotation Manual in between trials

Inter-annotator agreement

- Calculated Cohen's Kappa for all 8 classes
 - **Macro** - Arithmetic mean (each class contributes equally)
 - **Micro** - Weighted mean of classes (by how often either annotator said they were present)



- $\kappa \in [-1, 1]$
- Aim:
 - > 0.4 Moderate
 - > 0.6 Ideal

Inter-annotator agreement details

1. TOWIE C2

Confrontational
Multimodal

	SPos-	SNeg-	HPos-	HNeg-	SPos+	SNeg+	HPos+	HNeg+
Cohen	0.00	0.00	0.59	0.66	0.10	0.65	0.43	0.00
Count	4	11	28	32	33	36	43	9

2. WikiTalk

Confrontational
Unimodal

	SPos-	SNeg-	HPos-	HNeg-	SPos+	SNeg+	HPos+	HNeg+
Cohen	0.66	0.49	0.52	0.76	0.62	0.03	0.61	0.00
Count	2	8	25	13	8	10	26	9

3. TOWIE C4

Variable
Multimodal

	SPos-	SNeg-	HPos-	HNeg-	SPos+	SNeg+	HPos+	HNeg+
Cohen	0.58	NA	0.61	0.21	0.53	0.49	0.77	-0.04
Count	11	0	32	16	10	3	12	6

Challenges and Future Work

Mind reading

- Annotating takes a lot of mind reading.
- The annotator's perception (associations, biases, etc.) heavily determines how they reconstruct an interactant's internal state.
- Yet we need annotators to make these judgements on a case-by-case basis.

One effect, multiple turns

- Currently, we are annotating face effect for each turn
- But there can be a face effect as **a result of multiple turns**

A_{u1} : *What was the idea here?*

A_{u2} : *How does that help us?*

A_{u3} : *Why did you do that?*

u1, u2, u3 **individually**: requests for information (HNeg-)

u1-3 **together**: criticism (HPos-)

Annotation Manual: delineating intention vs effect

- The Annotation Manual only provides guidelines, not strict rules.
- The Annotation Manual provides examples of acts
 1. classified by their intended meaning and
 2. as conventionally having a specific effect on face.
- So this may not be ideal since we want to capture the *actual* (perlocutionary) effects on face.

Summing up

 We have developed an annotation system for face effects.

  **This tool can disambiguate the valence of the same utterance in different interactional contexts.**

   More! We can use it to explore other questions related to face in interaction.

 But it needs work (better agreement rate, weightiness of face effects incorporated, other challenges addressed).

 So that it can deal with more complex utterances (e.g., which have both face raising *and* threatening effects, which are part of a larger action, etc.).

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Annotation Manual



TOWIE Corpus

Thank you

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